Data Structures and Algorithms · 21CSC201J

Unit 3 · Session 9 · SLO 2 — Solved Worksheet

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Unit â€“ 3 : Session â€“ 9 : SLO â€“ 2 | Types of Queues â€” Match the Following

I. Answers

A. Linear Queue â€” 2. Basic FIFO structure

B. Circular Queue â€” 13. Last position connects to the first

C. Double-Ended Queue (Deque) â€” 11. Insertion and deletion at both ends

D. Input-Restricted Deque â€” 9. Deletion at both ends, insertion at one end only

• E. Output-Restricted Deque â€” 12. Insertion at both ends, deletion at one end only

• F. Priority Queue â€” 10. Elements served based on importance

• G. Simple Queue â€” 15. Insertion at rear, deletion at front

• H. Static Queue â€” 3. Fixed size queue implemented using arrays

• I. Dynamic Queue â€” 1. Grows or shrinks using linked list

• J. Circular Deque â€” 14. Ends are connected and operations at both ends

• K. Single-Ended Queue â€” 8. Operation allowed at only one end

• L. Blocking Queue â€” 5. Used for inter-thread communication

• M. Non-blocking Queue â€” 7. Doesn't wait for operations to finish

• N. Monotonic Queue â€” 6. Maintains increasing or decreasing order

• O. Queue of Queues â€” 4. Nested queue structure (queue as elements)